ALGEBRAIC-STACK AND DELIGNE-MUMFORD-STACK PROPERTIES ARE INVARIANT UNDER EQUIVALENCE OVER THE BIG FPPF SITE

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Editorial note. The source proves the forward implication and the DM variant in one pass; the converse is the identical symmetric rerun and is carried as an explicit obligation. Both same-chapter inputs are not carded ('Proof omitted.' and 'Omitted.') and are declared external.

Allowed standard prerequisites (editorial).

- Stack property under equivalence. Stacks, lemma stack-in-groupoids-equivalent (cross-chapter, named). Y is a stack in groupoids.
- Diagonal representability invariance. lemma-representable-by-spaces-morphism-equivalent (same chapter, NOT carded, declared external). Transfers representability of the diagonal along the equivalence square.
- Presentation property invariance. lemma-property-morphism-equivalent (same chapter, NOT carded, declared external). Transfers surjective smooth (resp. etale) along f.

Source reference. Original lemma 0459: sources/stacks-algebraic/pinned-original-source.tex, source lines 1090–1090. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0457: sources/stacks-algebraic/pinned-original-source.tex, source lines 802–802. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 042X: focused-seeds/geometry/S9-alg-equivalent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $\mathcal{X}$, $\mathcal{Y}$ be categories over $(Sch/S)_{fppf}$. Assume $\mathcal{X}$, $\mathcal{Y}$ are equivalent as categories over $(Sch/S)_{fppf}$. Then $\mathcal{X}$ is an algebraic stack if and only if $\mathcal{Y}$ is an algebraic stack. Similarly, $\mathcal{X}$ is a Deligne-Mumford stack if and only if $\mathcal{Y}$ is a Deligne-Mumford stack.*

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Proof. Assume $\mathcal{X}$ is an algebraic stack (resp. a Deligne-Mumford stack). By Stacks, Lemma 042X this implies that $\mathcal{Y}$ is a stack in groupoids over Sch_{fppf}. Choose an equivalence $f : \mathcal{X} \to \mathcal{Y}$ over Sch_{fppf}. This gives a 2-commutative diagram

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \\ {\scriptstyle \Delta_{\mathcal{X}}}\downarrow & & \downarrow{\scriptstyle \Delta_{\mathcal{Y}}} \\ \mathcal{X} \times \mathcal{X} & \xrightarrow{f \times f} & \mathcal{Y} \times \mathcal{Y} \end{array}$$

whose horizontal arrows are equivalences. This implies that $\Delta_{\mathcal{Y}}$ is representable by algebraic spaces according to Lemma 0457. Finally, let U be a scheme over S, and let $x : (Sch/U)_{fppf} \to \mathcal{X}$ be a 1-morphism which is surjective and smooth (resp. étale). Considering the diagram

$$\begin{array}{ccc} (Sch/U)_{fppf} & \xrightarrow{\text{id}} & (Sch/U)_{fppf} \\ {\scriptstyle x}\downarrow & & \downarrow{\scriptstyle f \circ x} \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array}$$

and applying Lemma 0459 we conclude that $f \circ x$ is surjective and smooth (resp. étale) as desired. □